The documentation and process conversion measures necessary to comply with this revision shall be completed by: 16 March 2020.

INCH-POUND

MIL-PRF-19500/586M
w/AMENDMENT 2
17 November 2019
SUPERSEDING
MIL-PRF-19500/586M
w/AMENDMENT 1
26 August 2019

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER, SCHOTTKY BARRIER, ENCAPSULATED (AXIAL LEADED AND SURFACE MOUNT) AND UNENCAPSULATED, TYPES 1N5817−1, 1N5819−1, AND 1N6761−1, QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device types as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for unencapsulated devices (die) as specified in MIL-PRF-19500.

1.2 Package outlines and die topography. The device packages for the encapsulated device types are as follows: Axial leaded DO−204AL (formerly DO−41) in accordance with figure 1 and metal electrode leadless face (MELF) DO-213AB in accordance with figure 2. The dimensions and topography for un-encapsulated die are in accordance with figure 3 for device types JANHC and JANKC.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	V_{RWM} (1)	$I_{O(PCB)}$ (2) T_A = +55°C	I_{FSM}	$R_{\theta JL}$ (3)	Max $R_{\theta JEC}$ (3)	Max $R_{\theta ja}$ (4)	T_J (5)	T_{STG}
	V (pk)	A dc	A dc	°C/W	°C/W	°C/W	°C	°C
1N5817−1	20	1.0	25	70		220	−65 to	
1N5817UR−1	20	1.0	25		40	220	+125	
1N5819−1	45	1.0	25	70		220	−65 to	−65 to
1N5819UR−1	45	1.0	25		40	220	+125	+150
1N6761−1	100	1.0	25	70		220	−65 to	
1N6761UR−1	100	1.0	25		40	220	+150	

(1) See figures 4, 5, 6, 7, 8, and 9 for derating curves and for effects of V_R on T_J.
(2) For both axial leaded and MELF (UR) mounted to a printed circuit board (PCB). See 6.7 for PCB application details and notes.
(3) For L = .375 inch (9.53 mm). For thermal impedance see figures 10 and 11.
(4) $R_{\theta JA}$ is measured at I_O = 1 A.
(5) The maximum T_J depends on the voltage applied.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Types	Max V_{F1} I_F = 0.1 A	Max V_{F2} I_F = 1.0 A	Max V_{F3} I_F = 3.1 A	Max I_{RM} at V_{RWM} pulsed method (see 4.5.1)		Max C_T V_R = 5 V dc
				T_J = +25°C I_{RM1}	T_J = +100°C I_{RM2}	
	V (pk)	V (pk)	V (pk)	µA	mA	pF
1N5817−1	.32	.45	.65	50	5.0	110
1N5817UR−1	.32	.45	.65	50	5.0	110
1N5819−1	.34	.49	.80	50	5.0	70
1N5819UR−1	.34	.49	.80	50	5.0	70
1N6761−1	.38	.69	NA	100	12.0	70
1N6761UR−1	.38	.69	NA	100	12.0	70

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs, and 6.8 for supersession information.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for un-encapsulated devices (die). The quality level designators for un-encapsulated devices that are applicable for this specification sheet are "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "5817", "5819", and "6761".

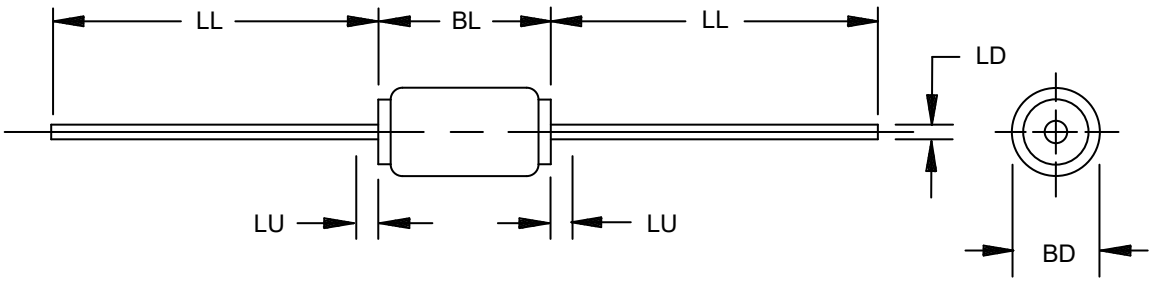
1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

−1	A dash-one (−1) designator identifies the package as an axial leaded DO-204AL that uses a metallurgically bonded double plug construction (see 3.4.3).
UR−1	A ur dash-one (UR−1) designator identifies the package is a surface mount DO-213AB that used a metallurgically bonded double plug construction (see 3.4.3).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.5 Die identifiers for unencapsulated devices. The manufacturer die identifiers that are applicable for this specification sheet are "A" and "B" (see figure 3 and 6.5.2).

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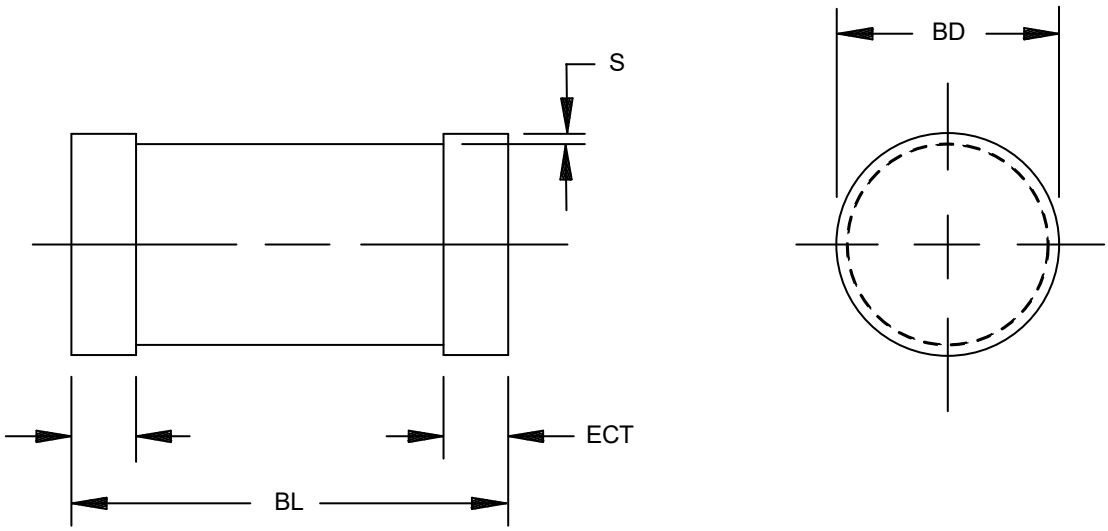
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.080	.107	2.03	2.72	3
BL	.160	.205	4.06	5.21	3
LD	.028	.034	0.71	0.86	
LL	1.000		25.40		
LU		.050		1.27	4

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Package contour optional within cylinder of diameter BD and length BL. Slugs, if any, shall not be included within this cylinder, but shall not be subject to the minimum limit of diameter BD.
4. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

Types 1N5817-1, 1N5819-1 and 1N6761-1

FIGURE 1. Physical dimensions for axial lead devices (DO-204AL, formerly DO-41).

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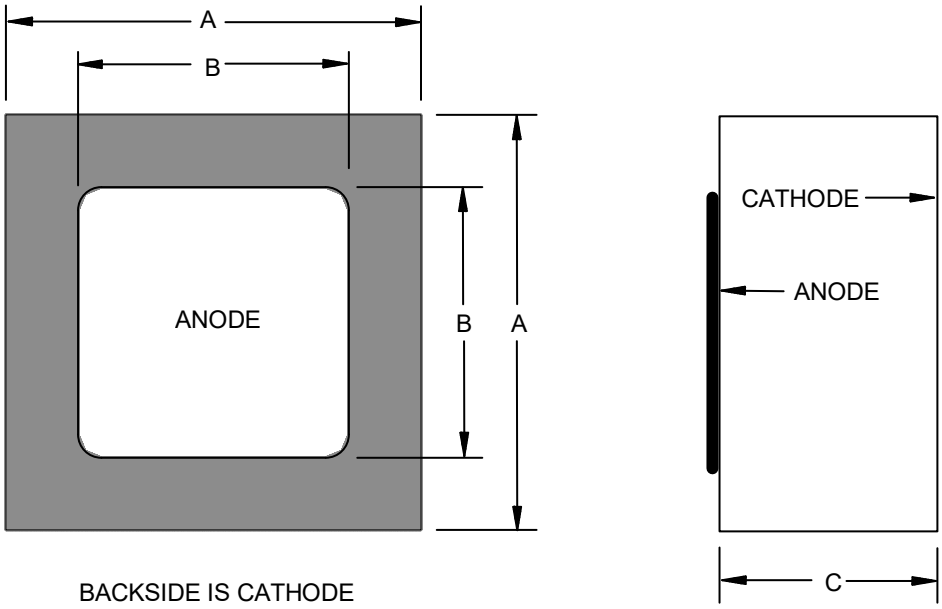
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.094	.105	2.39	2.67
BL	.189	.205	4.80	5.21
ECT	.016	.022	0.41	0.56
S	.001		0.03	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

Types 1N5817UR-1, 1N5819UR-1, and 1N6761UR-1

FIGURE 2. Physical dimensions for MELF devices (DO-213AB).

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Symbol	Dimensions for both A and B versions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.035	.039	0.89	0.99
B	.031	.033	0.79	0.84
C	.008	.012	0.20	0.31

Design data for both A and B versions	
Metallization:	
Top (anode):	Al
Back (cathode):	Au
Al thickness	25,000 Å minimum
Au thickness	4,000 Å minimum

FIGURE 3. Physical dimensions for un-encapsulated die (A and B version for JANHC and JANKC).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and herein. The device package styles are as follows: Axial leaded DO-204AL (formerly DO-41) in accordance with figure 1 for device types 1N5817-1, 1N5819-1, and 1N6761-1, DO-213AB (MELF) in accordance with figure 2 for device types 1N5817UR-1, 1N5819UR-1, and 1N6761UR-1, and unencapsulated die in accordance with figure 3 for device types JANHC and JANKC.

3.4.1 Lead finish. The lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2 and 6.8.2).

3.4.2 Polarity. The polarity of all packaged device types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UR suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used. The polarity of unencapsulated die shall be as identified on figure 3.

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3.4.3 Diode construction. All devices shall be metallurgically bonded, double plug construction in accordance with the requirements of MIL-PRF-19500. All glass diodes shall be designed with sufficient thermal compensation in the axial direction to optimize tensile and compressive stresses. Dimensional analysis is required of all materials used to achieve axial thermal compensation. Dimensional tolerances and corresponding coefficient of thermal expansion (CTE) shall be documented on the DLA Land and Maritime Design and Construction Form 36D and shall be approved by the qualifying activity to maintain qualification. Dimensional tolerances shall be sufficiently tight enough to prevent excessive stresses due to the inherent CTE mismatch. The UR suffix devices (DO–213AB) shall be structurally identical to the non-UR suffix devices except for end-cap lead attachment.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein. Initial container package marking shall be in accordance with MIL-PRF-19500.

3.5.1 Axial lead devices. Manufacturer's identification and date code shall be marked on the devices. The polarity shall be indicated with a contrasting color band to denote the cathode end. The prefixes JAN, JANTX, JANTXV, and JANS may be abbreviated as J, JX, JV, and JS respectively. The part number may be reduced to J5817, JX5817, JV5817, or JS5817. Color coding shall not be permitted for part numbering.

3.5.2 MELF surface mount devices (UR versions). For UR suffix devices only, all marking, except polarity may be omitted from the body, but shall be retained on the initial container.

3.5.3 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UR suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 Qualification of un-encapsulated die (JANHC and JANKC devices). Qualification inspection for unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. This qualification testing may be performed with the die mounted in a TO-5 package in lieu of the DO-204AL axial leaded package.

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4.3 Screening of packaged devices. Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTXV and JANTX level
2	Not required	Not required
3b 3c (1)	Not applicable Required (see 4.3.3)	Not applicable Required (see 4.3.3)
4, 5, and 6	Not applicable	Not applicable
9	I_{R1} and V_{F2}	Not applicable
10 (2)	T_A = +110°C; V_{RWM} = 20 V(pk), 1N5817, T_A = +110°C; V_{RWM} = 45 V(pk), 1N5819, T_A = +100°C; V_{RWM} = 100 V(pk), 1N6761, I_O = 0, V_R = V_{RWM}	T_A = +110°C; V_{RWM} = 20 V(pk), 1N5817, T_A = +110°C; V_{RWM} = 45 V(pk), 1N5819, T_A = +100°C; V_{RWM} = 100 V(pk), 1N6761, I_O = 0, V_R = V_{RWM}
11	ΔI_{R1} ≤ 100 percent of initial reading or .02 mA, whichever is greater. ΔV_{F2} ≤ ±50 mV dc	I_{R1} and V_{F2}
12	See 4.3.2	See 4.3.2
13	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial reading or .02 mA, whichever is greater; ΔV_{F2} ≤ ±50 mV dc	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial reading or .02 mA, whichever is greater; ΔV_{F2} ≤ ±50 mV dc

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with table E-IV of MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Test time shall be 48 hours minimum. T_J is not to exceed +115°C at V_{RWM}. T_J is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, T_J remains approximately equal to T_A or T_J for I_O = 0.

4.3.1 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements.

4.3.2 Burn-in conditions. The mounting and testing shall be in accordance with test condition B of method 1038 of MIL-STD-750. The burn-in conditions shall be as follows: I_F = 1.0 A dc (minimum), adjust I_F or T_A to achieve T_J = +100°C min.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} and t_{MD}. See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS), and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Condition
B3	1056	0°C to +100°C, 25 cycles.
B3	1051	−55°C to +150°C, 100 cycles.
B3	4066	I_{FSM} = 25 A (pk), condition A, I_O = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of t_p = 8.3 ms each at 1 minute intervals.
B3	2101	Decap analysis; scribe and break only.
B3	2075	In accordance with 4.5.3.
B4	1037	I_F = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.
B5	1026	I_F = 1 A dc (minimum), adjust I_F or T_A to achieve T_J = +125°C (minimum).
B6	4081	See 4.3.3; $R_{\theta JL}$ and $R_{\theta JEC}$ only.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	−55°C to +150°C, 25 cycles.
B2	4066	I_{FSM} = 25 A (pk), condition A, I_O = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of t_p = 8.3 ms each at 1 minute intervals.
B3	1027	I_F = 1 A dc (minimum), adjust I_F or T_A to achieve T_J = +125°C minimum.
B4	2101	Decap analysis; scribe and break only, see 4.5.3.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	0°C to +100°C, 10 cycles.
C2	1051	−55°C to +150°C, 25 cycles.
C2	2036	Tension: Test condition A; weight = 12 pounds (5.44 kg); t = 15 s. Lead fatigue: Test condition E; weight = 1 pound (0.45 kg). NOTE: Neither tension nor lead fatigue are applicable for UR devices.
C5	4081	See 4.3.3; $R_{\theta JL}$ and $R_{\theta JEC}$ only.
C6	1027	I_F = 1 A dc (minimum), adjust I_F or T_A to achieve T_J = +125°C minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with table E-IX of MIL-PRF-19500 and the conditions for subgroup testing in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each device under test still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through-out the burn-in period. Test method 3100 of MIL-STD-750 shall be used to measure T_J.

4.5.3 Decap internal visual scribe and break (DPA). Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die or bond material was in contact with the plug and verify metallurgical bonding area.

4.6 Element evaluation of un-encapsulated die. The element evaluation of un-encapsulated die shall be in accordance with appendix G of MIL-PRF-19500.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.3	$Z_{\theta JX}$			
Forward voltage	4011	Condition B; I_F = .1 A (pk), pulsed (see 4.5.1)	V_{F1}			
1N5817-1, 1N5817UR-1					.32	V dc
1N5819-1, 1N5819UR-1					.34	V dc
1N6761-1, 1N6761UR-1					.38	V dc
Forward voltage	4011	Condition B; I_F = 1.0 A (pk), pulsed (see 4.5.1)	V_{F2}			
1N5817-1, 1N5817UR-1					.45	V dc
1N5819-1, 1N5819UR-1					.49	V dc
1N6761-1, 1N6761UR-1					.69	V dc
Forward voltage	4011	Condition B; I_{FM} = 3.1 A (pk), pulsed (see 4.5.1)	V_{F3}			
1N5817-1, 1N5817UR-1					.65	V dc
1N5819-1, 1N5819UR-1					.80	V dc
Reverse current leakage	4016	DC or equivalent pulse method	I_{R1}			
1N5817-1, 1N5817UR-1		V_{RM} = 20 V			70	µA
1N5819-1, 1N5819UR-1		V_{RM} = 45 V			50	µA
1N6761-1, 1N6761UR-1		V_{RM} = 100 V			100	µA
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC or equivalent pulse method				
1N5817-1, 1N5817UR-1		V_{RM} = 20 V	I_{R2}		5.0	mA
		V_{RM} = 12 V	I_{R3}		3.0	mA
		V_{RM} = 6 V	I_{R4}		2.0	mA
1N5819-1, 1N5819UR-1		V_{RM} = 45 V	I_{R2}		5.0	mA
		V_{RM} = 40 V	I_{R3}		4.5	mA
		V_{RM} = 35 V	I_{R4}		4.0	mA
		V_{RM} = 24 V	I_{R5}		3.5	mA
		V_{RM} = 12 V	I_{R6}		2.5	mA
		V_{RM} = 6 V	I_{R7}		2.0	mA

See footnotes at end of table.

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TABLE I. Group A inspection − Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3 - Continued						
Reverse current leakage 1N6761-1, 1N6761UR-1	4016	DC or equivalent pulse method V_{RM} = 100 V V_{RM} = 50 V V_{RM} = 25 V V_{RM} = 12 V	 I_{R2} I_{R3} I_{R4} I_{R5}		 12.0 5.0 3.5 2.5	 mA mA mA mA
Forward voltage	4011	Condition B; I_F = 1.0 A (pk), pulse method (see 4.5.1)	V_{F4}			
1N5817-1, 1N5817UR-1 1N5819-1, 1N5819UR-1 1N6761-1, 1N6761UR-1					.40 .45 .64	V dc V dc V dc
Low temperature operation:		T_A = −55°C				
Forward voltage	4011	Condition B; I_F = 1.0 A (pk), pulse method (see 4.5.1)	V_{F5}			
1N5817-1, 1N5817UR-1 1N5819-1, 1N5819UR-1 1N6761-1, 1N6761UR-1					.60 .65 .80	V dc V dc V dc
Reverse current leakage 1N5817-1, 1N5817UR-1 1N5819-1, 1N5819UR-1 1N6761-1, 1N6761UR-1	4016	DC or equivalent pulse method V_{RM} = 20 V V_{RM} = 45 V V_{RM} = 90 V	 I_R		 100 200 400	 µA µA µA
Subgroup 4						
Capacitance	4001	V_R = 5 V dc, .01 ≤ f ≤ 1 MHz, V_{SIG} = 15 mV p-p	C_T			
1N5817-1, 1N5817UR-1 1N5819-1, 1N5819UR-1 1N6761-1, 1N6761UR-1					110 70 70	pF pF pF

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics for all UR devices are identical to the corresponding axial leaded devices unless otherwise specified.

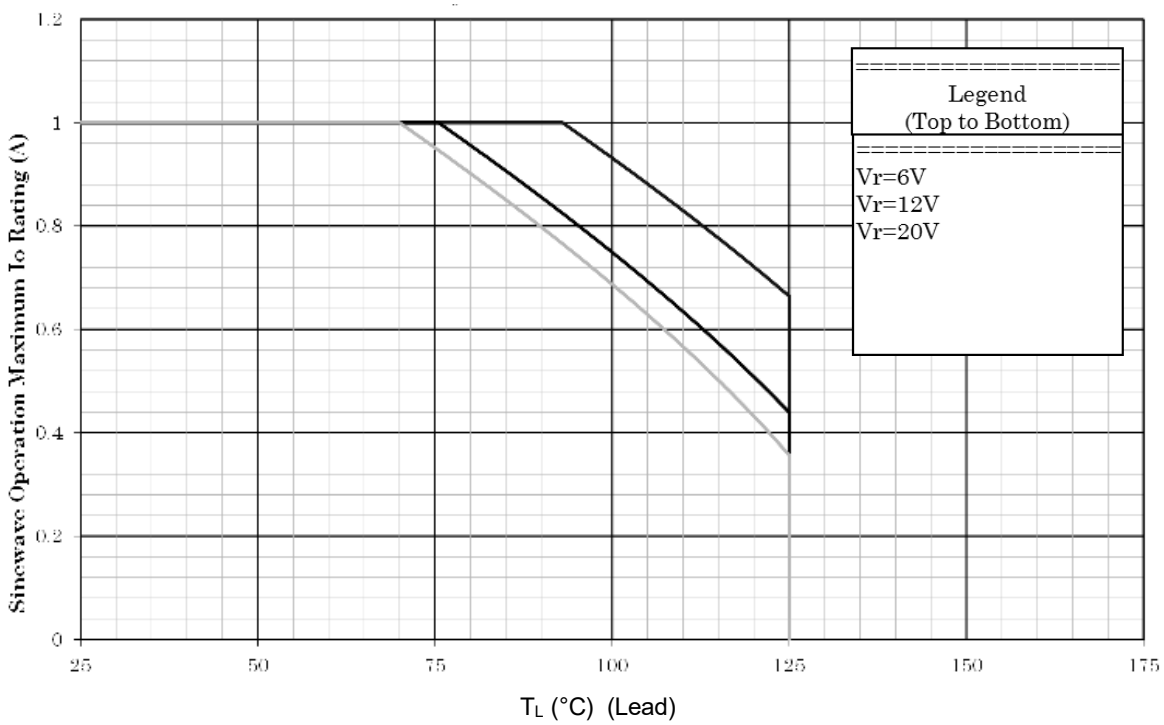
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TABLE II. Group E inspection (all quality levels) for qualification and re-qualification.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock	1056	100 cycles, 0°C to 100°C	
Temperature cycling	1051	500 cycles, −65°C to 150°C	
Hermetic seal	1071	Test condition E	
Electrical measurement		See table I, subgroup 2	
Subgroup 2			45 devices, c = 0
Intermittent operating life	1036	10,000 cycles, I_f = 1 A dc, T_{on} = T_{off} = 1 minute	
Electrical measurement		See table I, subgroup 2	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500	
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices, c = 0
Resistance to glass cracking	1057	Test condition B. Test to destruction or 25 cycles maximum, whichever comes first	

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Temperature - current derating curve family curves for 1N5817-1 at T_C = 25°C



Sinewave operation 50 percent duty cycle
$R_{\theta JL}$ = 70°C/W

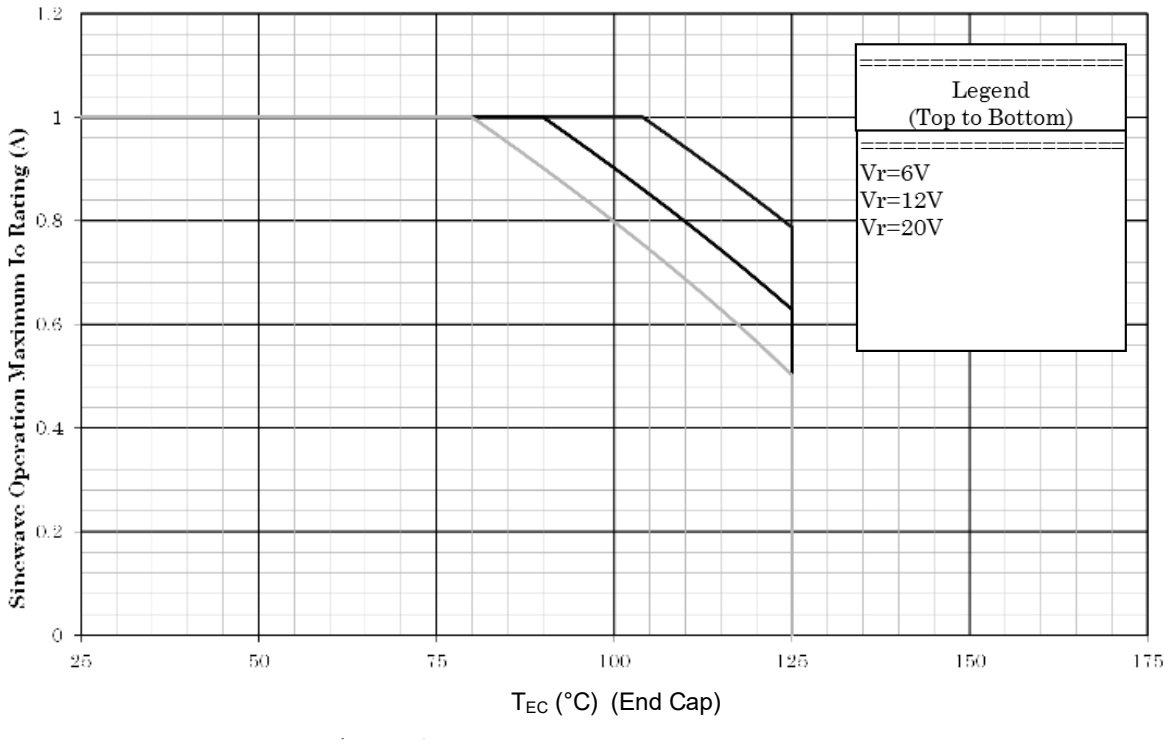
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.
3. Applies to 1N5817-1 up to 20V max.

FIGURE 4. Derating for 1N5817-1 (DO-204AL).

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Temperature - current derating curve family curves for 1N5817UR-1 at T_C = 25°C



Sinewave operation 50 percent duty cycle
$R_{\theta JL}$ = 40°C/W

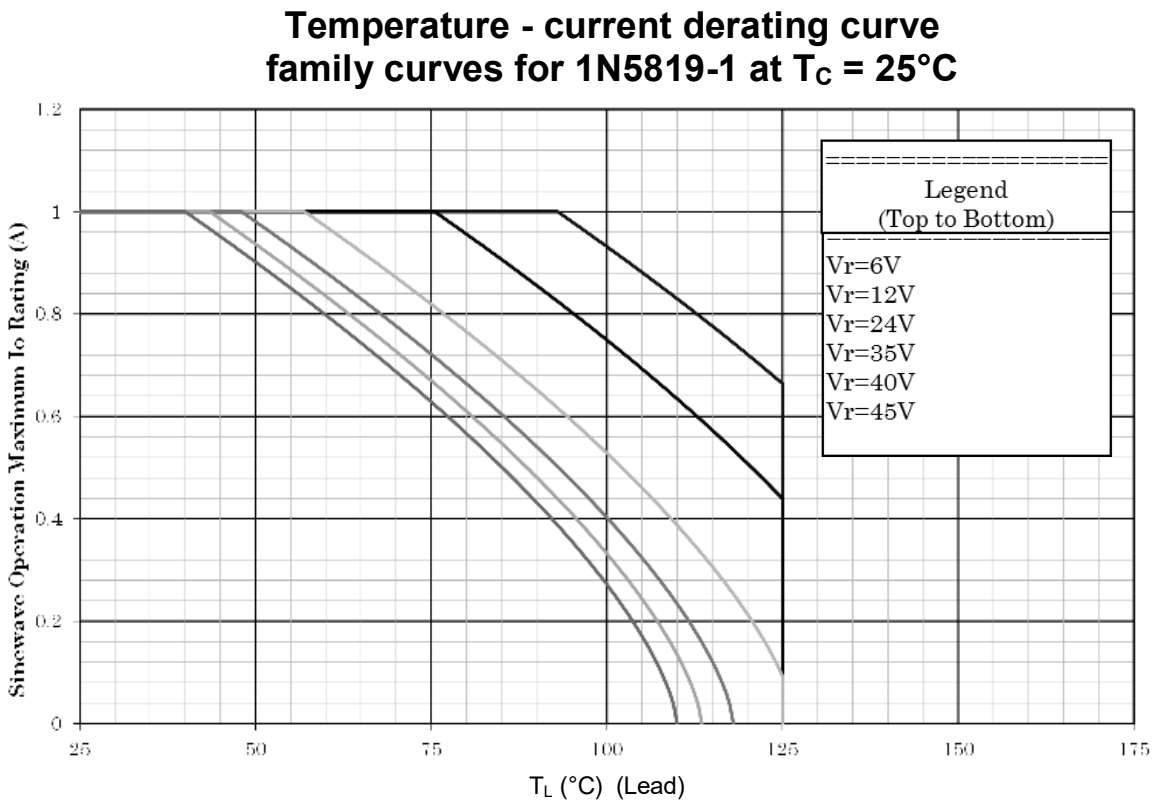
The curve for V_R = 6V coincides with the curve for V_R = 12V.

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.
3. Applies to 1N5817UR-1 only up to 20V max.

FIGURE 5. Derating for 1N5817UR-1 (DO-213AB).

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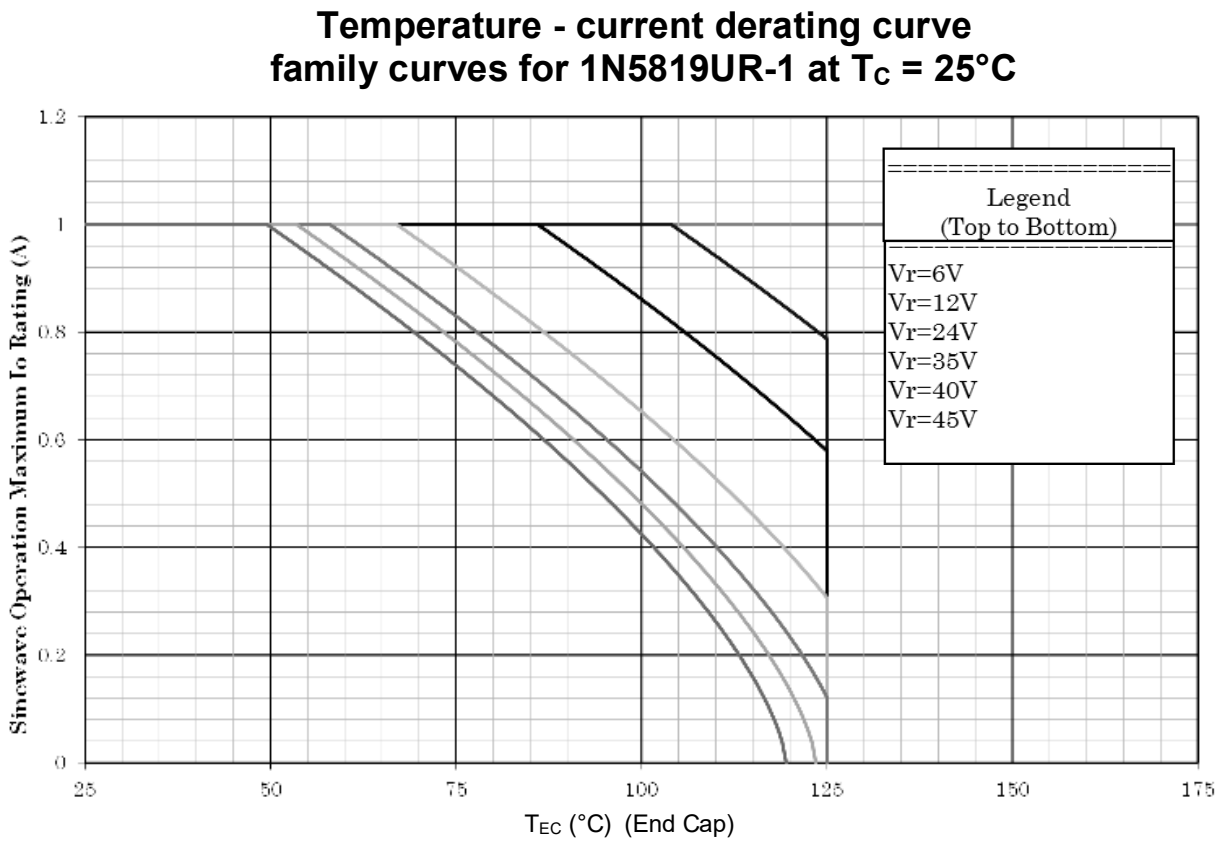
Sinewave operation 50 percent duty cycle
$R_{\theta JL}$ = 70°C/W

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 6. Derating for 1N5819-1 (DO-204AL).

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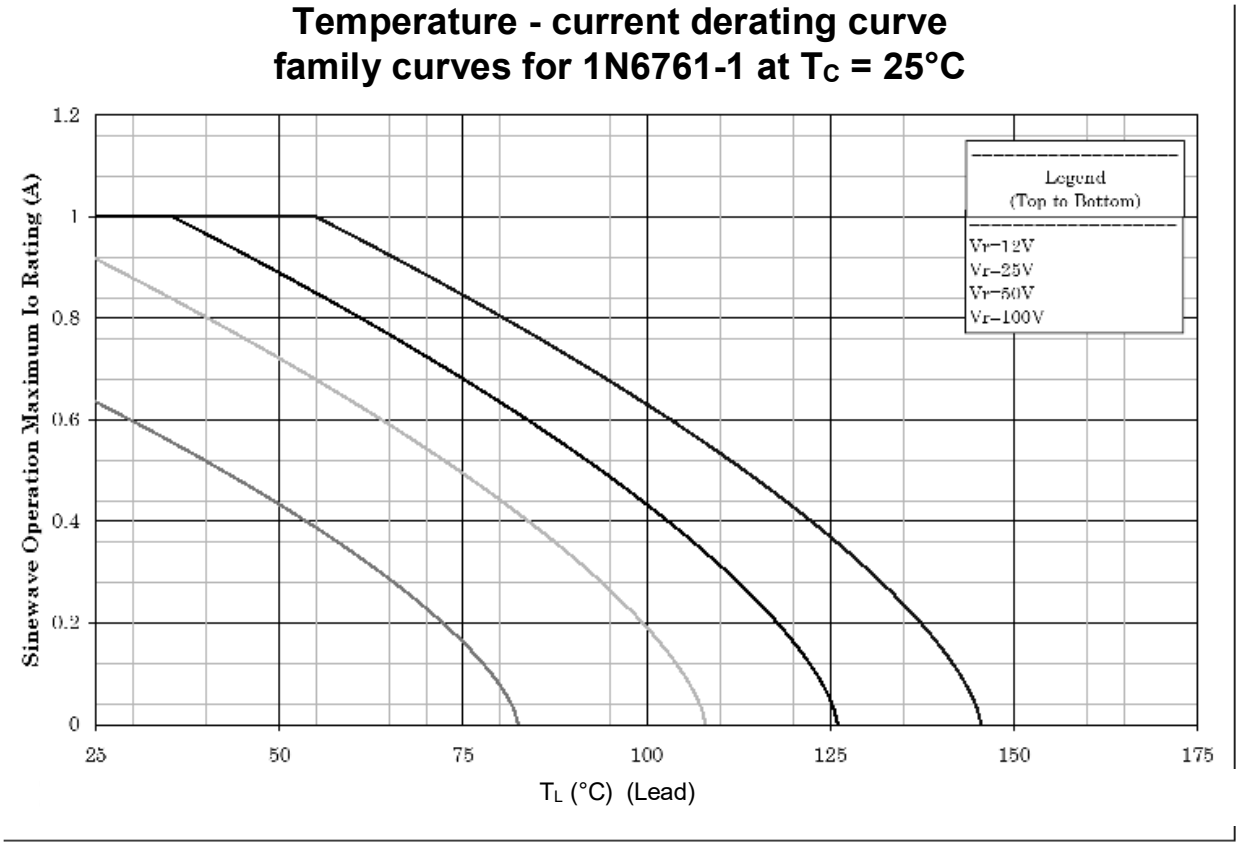
Sinewave operation 50 percent duty cycle
$R_{\theta JL}$ = 40°C/W

The curve for V_R = 6V coincides with the curve for V_R = 12V.

NOTES:
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 7. Derating for 1N5819UR-1 (DO-213AB).

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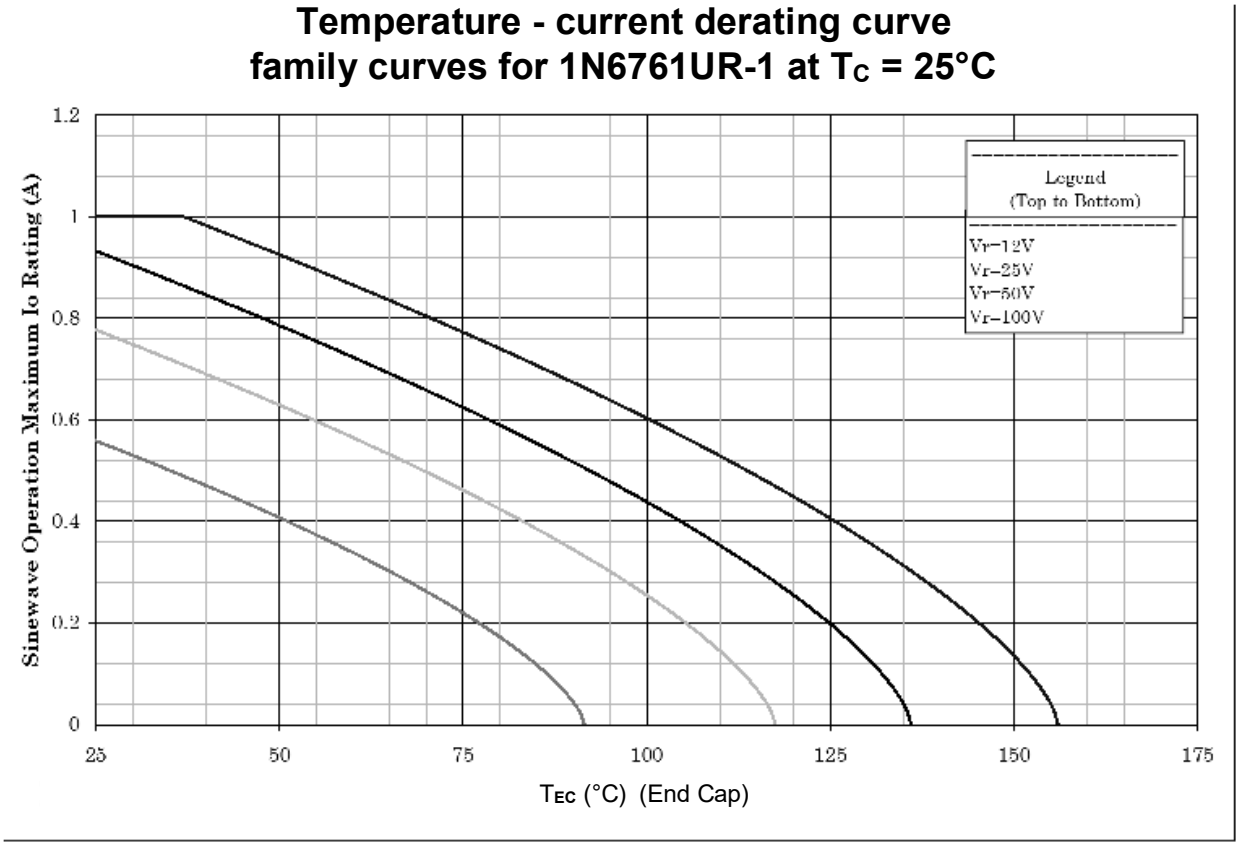


Sinewave operation 50 percent duty cycle
$R_{\theta JL}$ = 70°C/W

NOTES:
1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 8. Derating for 1N6761-1 (DO-204AL).

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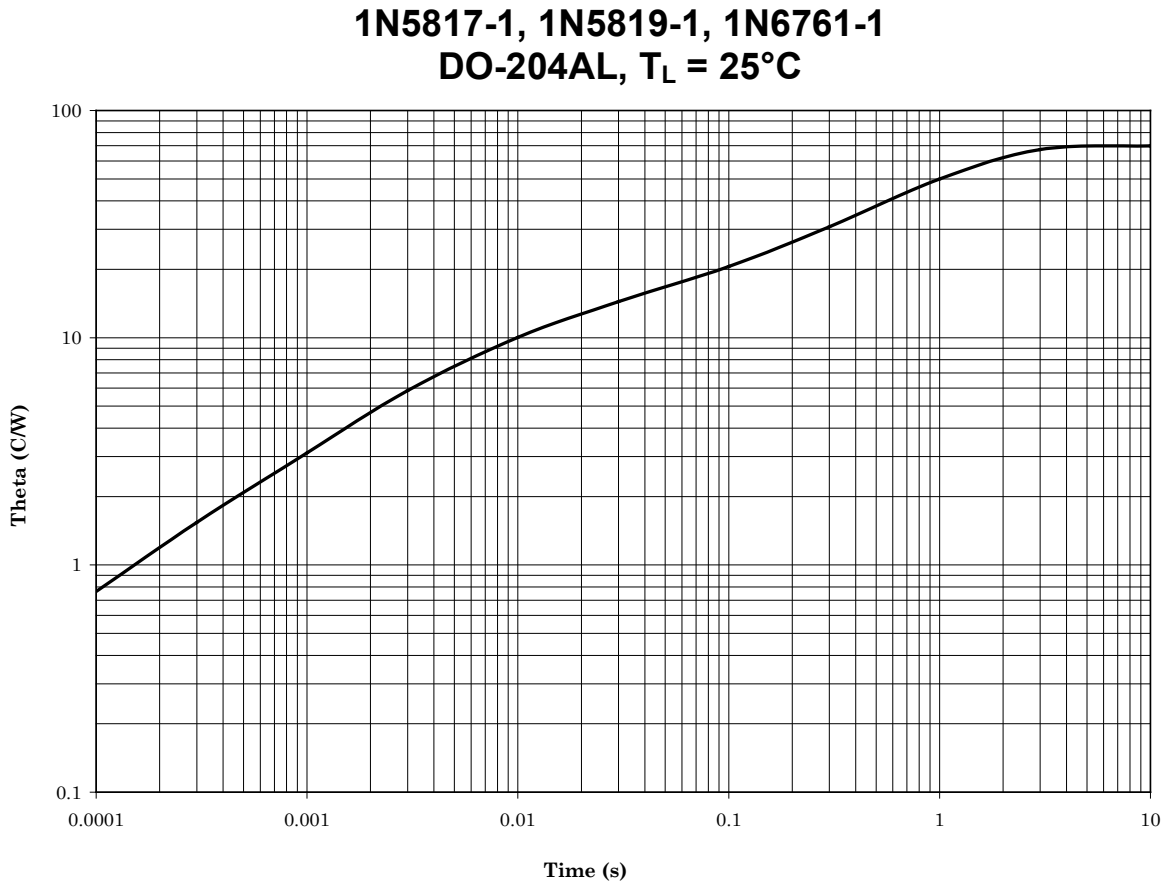
Sinewave operation 50 percent duty cycle
$R_{\theta JEC}$ = 40°C/W

NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. This temperature-current derating curve varies with applied voltage.

FIGURE 9. Derating for 1N6761UR-1 (DO-213AB).

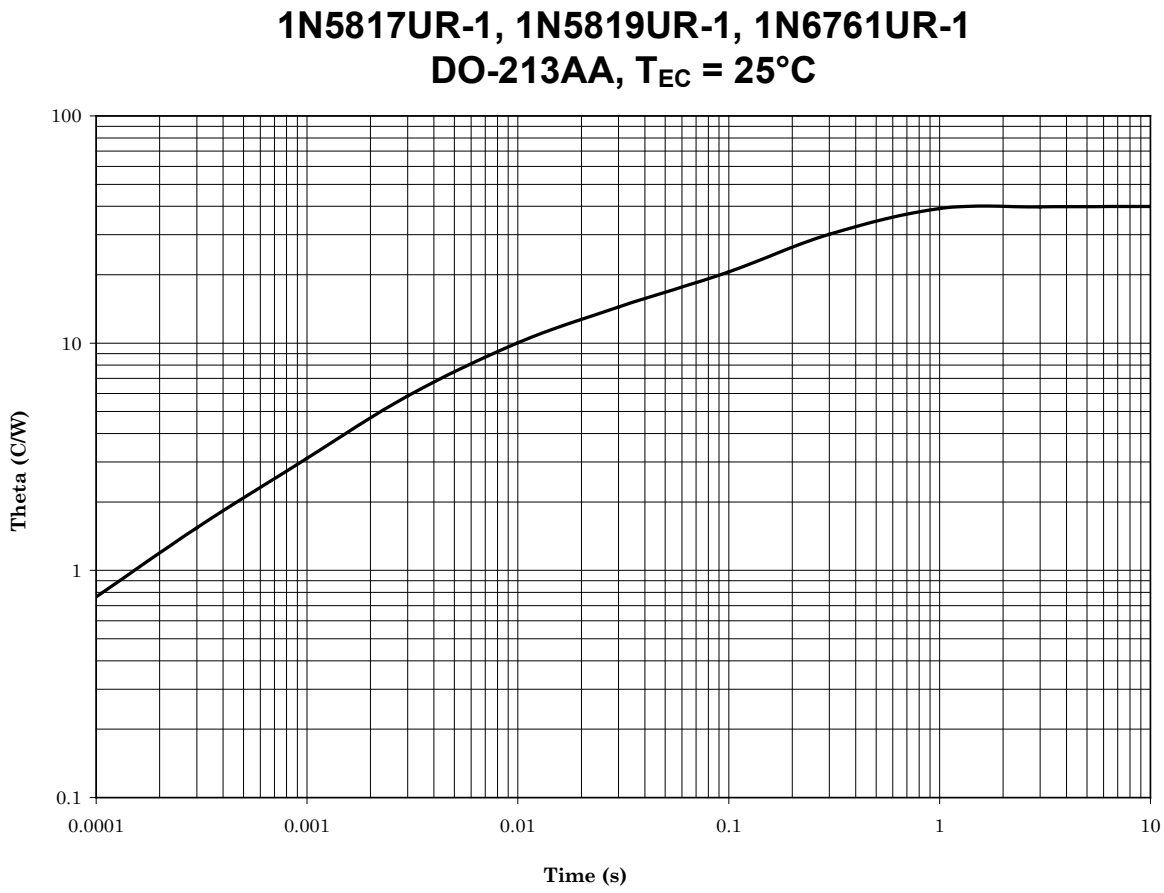
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L = .375 inch (9.53 mm).
$R_{\theta JL}$ = 70°C/W.
$Z_{\theta JX}$ = 10°C/W maximum at t_H = 10ms.

FIGURE 10. Thermal impedance for 1N5817-1, 1N5819-1 and 1N6761-1 (DO-204AL).

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L = 0.
$R_{\theta JEC}$ = 40°C/W.
$Z_{\theta JX}$ = 10°C/W maximum at t_H = 10ms.

FIGURE 11. Thermal impedance for 1N5817UR-1, 1N5819UR-1 and 1N6761UR-1 (DO-213AB).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Lead finish (see 3.4.1).

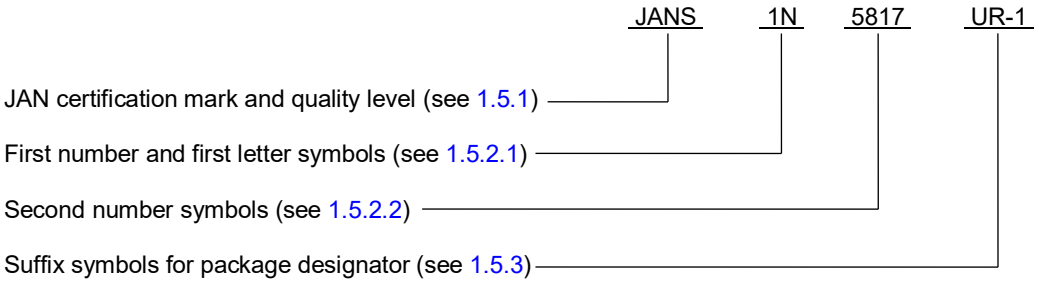
c. Packaging requirements (see 5.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QPDSIS 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

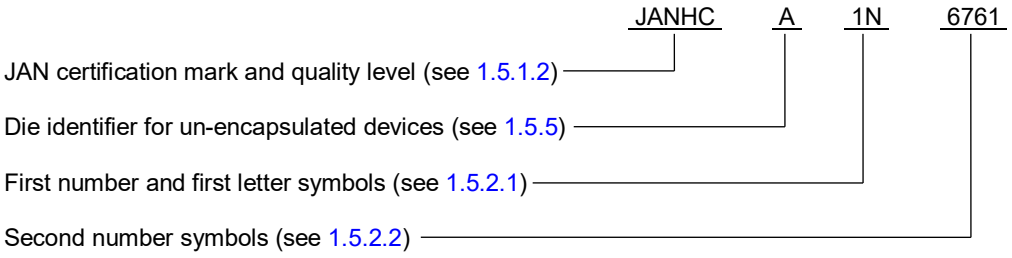
6.4 PIN construction examples. The PINs for encapsulated and un-encapsulated devices and are constructed using the following forms.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 Encapsulated devices. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

Quality level "JAN"	Quality level "JANTX"	Quality level "JANTXV"	Quality level "JANS"
JAN1N5817-1	JANTX1N5817-1	JANTXV1N5817-1	JANS1N5817-1
JAN1N5819-1	JANTX1N5819-1	JANTXV1N5819-1	JANS1N5819-1
JAN1N6761-1	JANTX1N6761-1	JANTXV1N6761-1	JANS1N6761-1
JAN1N5817UR-1	JANTX1N5817UR-1	JANTXV1N5817UR-1	JANS1N5817UR-1
JAN1N5819UR-1	JANTX1N5819UR-1	JANTXV1N5819UR-1	JANS1N5819UR-1
JAN1N6761UR-1	JANTX1N6761UR-1	JANTXV1N6761UR-1	JANS1N6761UR-1

6.5.2 Un-encapsulated devices. The following is a list of possible PINs available for un-encapsulated devices covered by this specification sheet.

Quality level "JANHC"		Quality level "JANKC"	
JANHCA1N5817	JANHCB1N5817	JANKCA1N5817	JANKCB1N5817
JANHCA1N5819	JANHCB1N5819	JANKCA1N5819	JANKCB1N5819
JANHCA1N6761	JANHCB1N6761	JANKCA1N6761	JANKCB1N6761

6.6 Suppliers of die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA1N5819) will be identified on the QPD.

Type	Manufacturer	
	43611	52GC4
1N5817	JANHCA1N5817 JANKCA1N5817	JANHCB1N5817 JANKCB1N5817
1N5819	JANHCA1N5819 JANKCA1N5819	JANHCB1N5819 JANKCB1N5819
1N6761	JANHCA1N6761 JANKCA1N6761	JANHCB1N6761 JANKCB1N6761

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6.7 Applications data. The $I_{O(PCB)}$ rating listed in 1.3 is typical for PCBs where the thermal resistance from the device mounting point to ambient is sufficiently controlled and where the maximum T_J listed in 1.3 is not exceeded. The PCB should be constructed using FR-4 base material with the following attributes: Base material thickness should not be less than .0625 inch (1.59 mm) in thickness; Cladding should be single sided copper foil with a weight of 1-ounce/square foot. The following pad sizes should be used for the appropriate package to ensure that a maximum $R_{\theta JA}$ of 220°C/W can be obtained when the device is mounted to a PCB with the following details.

6.7.1 Axial leaded devices. The pads for axial leaded devices should be as follows:

a. For through hole mounting the pad diameter should be .092 inch (2.34 mm) minimum.

b. For strip mounting the pad size should be .030 inch (0.76 mm) wide by 1 inch (25.4 mm) long. The lead length for mounting should be .187 inch (4.76 mm) or less from body to entry or contact point on PCB surface.

6.7.2 Surface mount MELF (UR) devices. The pads for UR devices should be .061 inch (1.55 mm) x .105 inch (2.67 mm).

6.8 Supersession information and obsoleted PINs.

6.8.1 Obsoleted PINs. Device type 1N6650 was removed from this specification sheet with the issuance MIL-PRF-19500/586F (17 October 2013). No supersession information is available for this device type.

6.8.2 Lead material. The original issue of this specification through MIL-PRF-19500/586L (26 March 2014) specified a lead material. Because of the performance format of this document, a lead material is no longer specified.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218−3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692−6939 or DSN 850−6939.

6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
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Air Force − 19

Preparing activity:
DLA − CC

(Project 5961−2019−127)

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